

Yonatan Sverdlov

Final-stage PhD Candidate in Mathematics, Technion (graduating 2026) | Seeking Research Internship / Research Scientist roles

Graph Neural Networks | Geometric Deep Learning | Weight-Space Learning

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Education

Technion – Israel Institute of Technology

PhD in Mathematics

Nov 2023 – Present

Advisor: Prof. Nadav Dym

Weizmann Institute of Science

MSc in Mathematics and Computer Science, GPA: 93.5

Mar 2021 – Jul 2023

Advisor: Prof. Shimon Ullman

Tel Aviv University

BSc in Mathematics and Computer Science, GPA: 92

Oct 2017 – Dec 2020

Research Interests

Graph neural networks; geometric deep learning; learning on model and weight spaces; expressivity and over-squashing; invariance and equivariance; bi-Lipschitz and stable representations.

Publications

- **Yonatan Sverdlov**, Benjamin Friedman, Snir Hordan, Nadav Dym. *When and How to Canonize: A Generalization Perspective*. **NeurIPS**, 2026.
- Soutrik Sarangi, **Yonatan Sverdlov**, Nadav Dym, Abir De. *Monotone and Separable Set Functions: Characterizations and Neural Models*. **NeurIPS**, 2025.
- Yaaqov Mishayev, **Yonatan Sverdlov**, Tal Amir, Nadav Dym. *Short-Range Oversquashing*. **LoG (Oral)**, 2025.
- **Yonatan Sverdlov**, Yair Davidson, Nadav Dym, Tal Amir. *FSW-GNN: A Bi-Lipschitz WL-Equivalent Graph Neural Network*. **LoG**, 2025.
- **Yonatan Sverdlov**, Nadav Dym. *On the Expressive Power of Sparse Geometric MPNNs*. **ICLR**, 2025.
- **Yonatan Sverdlov**, Ido Springer, Nadav Dym. *Revisiting Multi-Permutation Equivariance through the Lens of Irreducible Representations*. **ICLR**, 2025.

Preprints and Manuscripts

- Soutrik Sarangi, **Yonatan Sverdlov**, Adir Dayan, Haggai Maron, Nadav Dym. *Finite Probes Suffice: Identifiability and Universality for Weight-Space Learning*. **arXiv preprint; under review**, 2026.
- **Yonatan Sverdlov**, Eitan Rosen, Nadav Dym. *Toward Bi-Lipschitz Geometric Models*. **arXiv preprint**, 2025.
- **Yonatan Sverdlov**, Shimon Ullman. *Efficient Rehearsal Free Zero Forgetting Continual Learning using Adaptive Weight Modulation*. **arXiv preprint**, 2023.

Honors and Awards

Best Paper Award, Learning on Graphs Conference (LoG)

2025

Research Experience

Technion – Israel Institute of Technology

PhD Researcher

Nov 2023 – Present

Graph Neural Networks and Geometric Deep Learning

- Develop theoretical and empirical methods for graph, geometric, and weight-space learning, with emphasis on expressivity, oversquashing, stability, and symmetry.
- Design and implement PyTorch and PyTorch Geometric experiments to validate theoretical results and compare architectures across graph and model-space benchmarks.

Weizmann Institute of Science

MSc Researcher

Mar 2021 – Jul 2023

Computer Vision and Continual Learning

- Developed adaptive weight modulation for rehearsal-free continual learning while preserving performance on previously learned tasks.
- Implemented and evaluated deep-vision experiments across multiple continual-learning benchmarks.

Evrideo

AI Consultant

Aug 2023 – Oct 2023

Applied Computer Vision

- Built a computer-vision proof of concept to evaluate the feasibility of an early-stage ML product.

Technical Skills

Programming

Python, Java, C++

ML / Scientific

PyTorch, PyTorch Geometric, NumPy

Tools

Git, GitHub, LaTeX

Languages

Hebrew (native), English (professional proficiency)